

Week 4 Assignment

Repository Link:

<https://github.com/Hughes-Tyler/AnalyticsCapstone>

Report Link:

<https://github.com/Hughes-Tyler/AnalyticsCapstone/blob/main/notebooks/main.ipynb>

For this week's assignment, we started by looking back at the issues we flagged in week 3 and figuring out what to actually do about them. The first issue was that our calculations were being thrown off by a company with a funding date set in the year 2105 (Rasyonel is the company name). Obviously this was a typo in the data, but this one row caused our variable `funding_span_days` to display a value of over 33,000 days. This minor fix helped set that variable back on track. Next, we dropped 23 companies with no first funding date, since we can't actually calculate any of the timing features without that crucial piece.

Back in Week 3 we figured we'd just remove companies that were funded before they were founded, but there are about 3,700 of them and closed companies show up more in that group (12%) than in the full dataset (8%), so dropping them all would cut into one of the outcomes we're trying to predict. Over half the founding dates in the data are set to January 1st, which tells us a lot of them are just a year with a placeholder month and day. We only dropped the 100 that were funded more than five years before founding, then flagged the rest with a new `funded_before_founding` column and set their `negative_days_to_first_funding` to 0.

The last piece for the first part of our efforts was duplicate companies. Most of the 146 repeated names turned out to be different companies that just share a name (like two Amplitudes in different regions), so we only treated a row as a real duplicate if the name, founding date, and total funding all matched. This logic left us with 7 companies that were listed twice. For 3 of them (3DIVAZ, POLYGLOTS Inc., and PeopleEasy.com), one row said closed and the other said operating, and since there's no way to tell which is right, we dropped them both. For the other 4 the rows were identical, so we only kept one of each. All in, we went from 49,771 rows down to 49,638, plus one new column we can use as a feature when it comes time for modeling.

Once the row level cleaning was done, we moved on to the missing values that were left. A few companies had a total funding amount under \$100, which isn't realistic for a startup, so

we set those 16 to be classified as missing funding. About 18% of companies were missing funding, and the rate changes a lot by outcome (6.4% for IPO companies vs. 19.0% for closed ones), which tells us the missingness itself carries some signal. To keep that, we added a `funding_missing` column before filling anything in, then filled the gaps with the median (about \$1.85M). We went with the median over the mean because a handful of billion dollar companies drag the mean way up. We handled `country_code` and `category_list` the same way, since closed companies were missing those far more often (19.9% vs. 1.9% to 6.6% for country, and 8.3% vs. 2.3% or less for category). We filled both with "Unknown" instead of defaulting to USA, which would've been wrong about a third of the time and hidden the pattern. Last, we dropped 13 columns that were identifiers, raw versions of columns we'd already cleaned, or had too many unique values to be useful, like `homepage_url` with 47K unique values.

Next we handled the lopsided numeric columns. Funding total was the worst, skewed heavily because a few companies raised billions while most raised a few million or less. Funding span and days to first funding were also skewed pretty heavily. We applied a log transform to all three, using `log1p` so companies with a value of 0 don't break anything, and dropped the raw versions so the model doesn't see the same information twice. Funding went down to -0.24 and funding span went to 0.50, which is a big improvement from where we started. We also capped `funding_rounds` at 10, which only affected 110 companies (0.22%), since a simple count like that didn't need a log transform. We kept all 51 companies with \$1B or more in funding, like Uber and Facebook, because we decided that just because they ended up getting billions in funding, they are still real companies that can give us useful data for the IPO category, our smallest class.

The last step was turning the text columns into numbers so the models can use them. There are 127 countries in the data, and 86 of them have fewer than 50 companies, so one-hot encoding all of them would add a lot of columns that are almost always 0. Instead, we kept the 10 biggest countries (USA, GBR, CAN, IND, FRA, CHN, ISR, DEU, ESP, and AUS), which cover about 85% of companies, kept "Unknown" as its own group since closed companies are missing country more often, and put everything else in an "Other" group. That gave us 12 country columns. Each company can also have several category tags (2.6 on average and up to 44), and there are 23,130 different tag combinations, so regular one-hot encoding wouldn't work well. We split the tags apart and kept the 30 most common ones, which cover about 79.7% of companies, and made a 0/1 column for each tag, plus one for "Unknown" and one for companies with any tag outside the top 30. That gave us 32 category columns. We also

converted our target, the startup's status, to numbers (operating = 0, acquired = 1, closed = 2, ipo = 3) instead of one-hot encoding it, since the model needs a single column to predict. We then dropped the original country and category columns. We didn't do any text cleaning, since there is no free text in the data and the tags were already clean, with no whitespace problems or duplicate tags that only differed by case. No rows were removed, so we ended with 49,638 rows and 52 columns, with no text columns and no missing values.